

Concept Questions: Chapter 12

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ME-3640 – Heat Transfer

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Abstract

Fundamental principals and concepts are far more important to engineering expertise than understanding of example systems. These concept questions (shamelessly lifted from the textbook) will help you test your knowledge after reading the chapter summary and building a concept map. Use the relative ease and difficulty you experience while answering these questions to prioritize your efforts for the remainder of this chapter.

1 Electromagnetic and Thermal Radiation

- 12-1C** What is an electromagnetic wave? How does it differ from a sound wave?
- 12-2C** By what properties is an electromagnetic wave characterized? How are these properties related to each other?
- 12-3C** What is thermal radiation? How does it differ from the other forms of electromagnetic radiation?
- 12-4C** How does microwave cooking differ from conventional cooking?
- 12-5C** What is visible light? How does it differ from the other forms of electromagnetic radiation?
- 12-6C** What is the cause of color? Why do some objects appear blue to the eye while others appear red? Is the color of a surface at room temperature related to the radiation it emits?
- 12-7C** Why do skiers get sunburned so easily?
- 12-8C** How do ultraviolet and infrared radiation differ? Do you think your body emits any radiation in the ultraviolet range? Explain.
- 12-9C** Why is radiation usually treated as a surface phenomenon?

2 Blackbody Radiation

- 12-18C** What is a blackbody? Does a blackbody actually exist?
- 12-19C** Define the total and spectral blackbody emissive powers. How are they related to each other? How do they differ?
- 12-20C** Why did we define the blackbody radiation function? What does it represent? For what is it used?
- 12-21C** Consider two identical bodies, one at 1000 K and the other at 1500 K. Which body emits more radiation in the shorter-wavelength region? Which body emits more radiation at a wavelength of 20 mm?

3 Radiation Intensity

- 12-41C** What does a solid angle represent, and how does it differ from a plane angle? What is the value of a solid angle associated with a sphere?

- 12-42C** For a surface, how is irradiation defined? For diffusely incident radiation, how is irradiation on a surface related to the intensity of incident radiation?
- 12-43C** For a surface, how is radiosity defined? For diffusely emitting and reflecting surfaces, how is radiosity related to the intensities of emitted and reflected radiation?
- 12-44C** When the variation of spectral radiation quantity with wavelength is known, how is the corresponding total quantity determined?
- 12-45C** How is the intensity of emitted radiation defined? For a diffusely emitting surface, how is the emissive power related to the intensity of emitted radiation?

4 Radiation Properties

- 12-65C** What is a graybody? How does it differ from a blackbody? What is a diffuse gray surface?
- 12-66C** Define the properties emissivity and absorptivity. When are these two properties equal to each other?
- 12-67C** Define the properties reflectivity and transmissivity and discuss the different forms of reflection.
- 12-68C** What is the greenhouse effect? Why is it a matter of great concern among atmospheric scientists?
- 12-69C** We can see the inside of a microwave oven during operation through its glass door, which indicates that visible radiation is escaping the oven. Do you think that the harmful microwave radiation might also be escaping?

5 Atmospheric and Solar Radiation

- 12-88C** When the earth is closest to the sun, we have winter in the northern hemisphere. Explain why. Also explain why we have summer in the northern hemisphere when the earth is farthest away from the sun.
- 12-89C** Explain why surfaces usually have quite different absorptivities for solar radiation and for radiation originating from the surrounding bodies.
- 12-90C** You have probably noticed warning signs on the highways stating that bridges may be icy even when the roads are not. Explain how this can happen.

- 12-91C** What changes would you notice if the sun emitted radiation at an effective temperature of 2000 K instead of 5762 K?
- 12-92C** What is the solar constant? How is it used to determine the effective surface temperature of the sun? How would the value of the solar constant change if the distance between the earth and the sun doubled?
- 12-93C** Explain why the sky is blue and the sunset is yellow-orange.
- 12-94C** What is the effective sky temperature?

A Rubrics

Submissions of this assignment will be graded upon the criteria found in this section. Review these carefully, and complete your work so that application of this rubric produces your desired score.

 Criterion 1 Problems	
Homework is completed with appropriate effort, neatness, etc.	
Points	Description
20	Honest effort to participate and learn: every problem shows an answer or questions which need resolution.
10	Rushed or haphazard work: some problems show no effort at all.
0	Work is absent, wholly incorrect, or plagiarized.